

[illegible]

$$X \sim \text{Uniform}(0, 10)$$

$$Y \sim \text{Uniform}(2, 8)$$

$$E[X] = \sum_{k=0}^{10} k \cdot \frac{1}{11} \quad k=1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \quad E[Y] = \sum_{k=2}^8 \frac{1}{7} = \frac{35}{7} = 5$$

$$= \frac{55}{11} = 5$$

$$= \frac{55}{11} = 5$$

$$Z = X + Y$$

$$E[Z] = E[X + Y] = E[X] + E[Y] = 5 + 5 = 10$$

$$\text{Var}(X): E[X^2] = \sum_{k=0}^{10} k^2 \cdot \frac{1}{11} = \frac{385}{11} = 35$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = 35 - 5^2 = 10$$

$$\text{Var}(Z) = \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \Leftarrow X \text{ and } Y \text{ are independent of each other}$$

$$\text{Var}(Y) = E[Y^2] - E[Y]^2 = 29 - 25 = 4$$

$$\text{Var}(Z) = \text{Var}(X) + \text{Var}(Y) = 10 + 4 = 14$$

$$\text{Cov}(X, Z) \quad X=x \quad Z=x+Y$$

$$\text{Cov}(X, X+Y) = \text{Cov}(X, X) + \text{Cov}(X, Y) = E[XZ] - E[X]E[Z]$$

X and Y are independent of each other

$$\text{Cov}(X, X) = \text{Var}(X) = 10 \quad \text{Cov}(X, Y) = 0$$

$$\therefore \text{Cov}(X, Z) = 10$$

i) $(A - \lambda I)x = 0$

for λ_i is eigenvalue of A

$\det(A - \lambda I) = 0$

~~$\det(A - \lambda I) = 0$~~

Matrix(B) eigenvalue(B) $\det(B)$ $\text{tr}(B)$

A λ_i $\prod \lambda_i$ $\sum \lambda_i$

for suppose $P = [x_1 \dots x_n]$

A^k λ_i^k $\prod \lambda_i^k$ $\sum \lambda_i^k$

$A = P \Lambda P^{-1}$ $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$

A^{-1} λ_i^{-1} $\prod \lambda_i^{-1}$ $\sum \lambda_i^{-1}$

$A^k = (P \Lambda P^{-1})^k$

$= P \Lambda P^{-1} P \Lambda P^{-1} \dots P \Lambda P^{-1}$

$= P \Lambda^k P^{-1} (P^{-1} P = I)$

$Ax = \lambda x$ then $A^k x = A^{k-1}(Ax) = A^{k-1}(\lambda x) = \dots = \lambda^k x$

$x \neq 0 \Rightarrow \lambda \Rightarrow A^{-1}$

ii) $A = P \Lambda P^{-1}$ $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$

only one $x \neq 0$ $Ax = \lambda x$

$A^2 x = \lambda^2 x$

$(A + cI)x = (\lambda + c)x$

$A^{-1}x = \lambda^{-1}x$

$P(A)x = P(\lambda)x$

~~10~~

$$f(x; \theta) = \frac{1}{\theta+1} x^\theta$$

$$L(\theta) = \prod_{i=1}^n (\theta+1) x_i^\theta = (\theta+1)^n \prod_{i=1}^n x_i^\theta$$

$$\ell(\theta) = n \log(\theta+1) + \theta \sum_{i=1}^n \log x_i$$

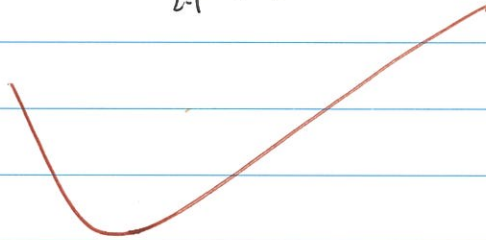
$$\text{suppose } S = \sum_{i=1}^n \log x_i \quad \frac{d\ell}{d\theta} = \frac{n}{\theta+1} + S = 0$$

$$\frac{n}{\theta+1} = -S$$

$$\theta+1 = -\frac{n}{S}$$

$$\theta = -\frac{n}{S} - 1$$

$$\hat{\theta} = -\frac{1}{\sum_{i=1}^n \log x_i} \cdot n - 1$$



题号

Question No.:

4

	Y=1	Y=2	Y=3	Y=4	Y=5		
X=1	0.04	0.04	0.04	0.04	0.04	$P(X=1) = 0.04 \times 5 = 0.2$	$P(Y=1) = 0.2$
X=2	0.08	0.04	0.08	0	0	$P(X=2) = 0.2$	$P(Y=2) = 0.2$
X=3	0.08	0	0.04	0.04	0.04	$P(X=3) = 0.2$	$P(Y=3) = 0.2$
X=4	0	0.12	0	0.08	0	$P(X=4) = 0.2$	$P(Y=4) = 0.2$
X=5	0	0	0.04	0.04	0.12	$P(X=5) = 0.2$	$P(Y=5) = 0.2$

$$H(X) = -\sum_x P(X=x) \log P(X=x)$$

$$= -[0.2 \log 0.2 + 0.2 \log 0.2 + 0.2 \log 0.2 + 0.2 \log 0.2 + 0.2 \log 0.2]$$

$$= -\log 0.2 = -\log_5 \frac{1}{5} = 1 \text{ bit}$$

$$H(Y) = -\sum_y P(Y=y) \log P(Y=y) = -\log 0.2 = 1 \text{ bit}$$

$$H(X, Y) = -\sum_x \sum_y P_{X,Y}(x, y) \log P_{X,Y}(x, y)$$

$$= -[5 \log 0.04 + 8 \log 0.08 + 6 \log 0.12] = -[5 \log_5 \frac{1}{25} + 8 \log_5 \frac{1}{125} + 6 \log_5 \frac{1}{75}]$$

$$= -[10 + 16 + 12] = -38$$

$$H(X|Y) = H(X, Y) - H(Y) = -38 - 1 = -39$$

$$H(Y|X) = H(X, Y) - H(X) = -38 - 1 = -39$$

$$I(X; Y) = H(X) - H(X|Y) = 1 - (-39) = 40$$

$$I(X; Y) = H(Y) - H(Y|X) = 1 - (-39) = 40$$

题号

Question No.: 5

(i) $z = A = 2$ (office) $N = 2 + 2 = 4$

$B = 2$ (home)

$p_B = 0.2 = \frac{1}{5}$ $q_B = 0.8 = \frac{4}{5}$ $\frac{q_B}{p_B} = 4$

$p \neq q$ $W_B = \frac{1 - (\frac{q}{p})^2}{1 - (\frac{q}{p})^4} = \frac{1 - (4)^2}{1 - (4)^4} = \frac{1}{17}$

$E_B = \frac{2}{q - p_B} - \frac{4}{q - p_B} \cdot \frac{1 - (\frac{q}{p_B})^2}{1 - (\frac{q}{p_B})^4}$ $q - p_B = 0.8 - 0.2 = 0.6$

$= \frac{2}{0.6} - \frac{4}{0.6} \cdot \frac{1}{17} = \frac{50}{17}$

(ii)

X

-10

(ii) day 1

① morning = m	m_{rain}	e_{rain}	$0.2 \times 0.8 = 0.16$
evening = e	m_{hot}	e_{hot}	$0.8 \times 0.8 = 0.64$
	m_{hot}	e_{rain}	$0.8 \times 0.2 = 0.16$

day 2

$W_1 = \frac{1 - (\frac{q}{p})}{1 - (\frac{q}{p})^4} = \frac{3}{255} \cdot \frac{4}{25}$

$W_2 = \frac{1}{17} - \frac{4}{25}$

$W_3 = \frac{1 - (\frac{q}{p})^3}{1 - (\frac{q}{p})^4} = \frac{63}{255} \cdot \frac{4}{25}$

答题不得写在红线外

Do not write your answer outside the red margins.

~~$\frac{a+b}{n} \rightarrow$~~

let $a \rightarrow 1$

$$\lim_{a \rightarrow 1} P_n^{(\alpha, \beta)} = \left\{ 1 - \left(\frac{\log a}{\log n} \right)^\beta, \frac{1}{T} \left(\frac{\log a}{\log n} \right)^\beta, 0, 0, 0 \right\}$$

$$= \{1, 0, 0, 0, \dots, 0\}$$

X

10